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Roll No

CE-8004(4)-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Finite Element Method

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Write any four differences between finite difference method and finite element method.
b) Write about the preprocessing and post processing in finite element method.
2. Taking the example of a single element cantilever Beam show that single point Gaussian Integration overcomes the problem of Shear Locking.
3. State and explain the convergence requirements of Polynomial Shape function.
4. Use Lagrange functions to find and draw the shape functions of a three noded triangular element.

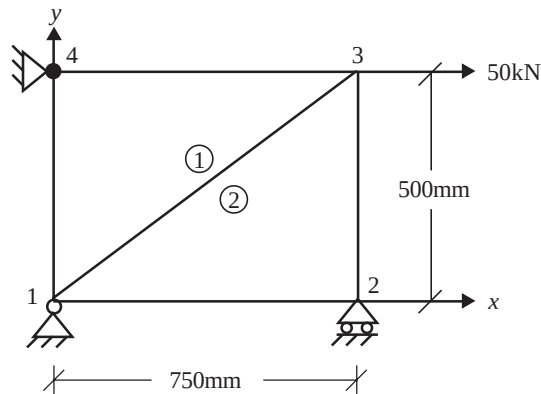
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5. Find the nodal displacement and element stresses in the propped beam shown below. Idealize the beam into two CST elements as shown. Assume plane stress condition.

Take $\mu = 0.25$ $E = 2 \times 10^5 \text{ N/mm}^2$, Thickness = 15 mm.



6. Explain the following terms:
- a) Nodes
 - b) Primary Nodes
 - c) Secondary Nodes
 - d) Internal Nodes
7. Solve the following equations by Gaussians eliminations method:

$$3X_1 + X_2 + X_3 = 4$$

$$X_1 + 4X_2 - X_3 = -5$$

$$X_1 + X_2 - 6X_3 = -12$$

8. Explain the following terms clearly:
- i) Local coordinates
 - ii) Global coordinates
 - iii) Natural coordinates
 - iv) Area coordinates

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